Lightweight OCR in North Sámi Heritage Digitization for Smart Learning

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***Abstract*—North Sámi is an endangered Uralic language, facing critical challenges in written heritage preservation. State-of-the-art Transformer-based OCR (Optical Character Recognition) models get 91.45% accuracy, but need 350 MB+ storage and over 12 seconds for inference on a single image on a CPU, which limits their practicality, especially in resource-constrained digitization. In this research, we propose a model comprising a lightweight integrated pipeline of a CNN-CTC (Convolutional Neural Network and Connectionist Temporal Classification) architecture to conduct OCR of Sámi language texts with resource-aware transcription. We offer systematic evaluation of our proposed CNN-CTC model (comparing it to 7 TrOCR variants from National Library of Norway), testing on 1,048 North Sámi text samples, including synthetically rendered historical content in Johan Turi's 1910 work. Our CNN-CTC model gives 87.76% character accuracy, trailing the best Transformer by 3.69 pp in CER, while operating 380 times faster, taking only 23 MB, enabling throughput of 109,000 images per hour compared to 290 for Transformers. We observe that the CNN-CTC model maintains stable accuracy when tested on synthetically rendered historical text where a transformer model degrades by 4.84 pp. Additionally, we demonstrate that linguistically motivated conjunction-based sentence splitting improves long sentence accuracy by 7.05 pp as a proof of concept for preprocessing-based mitigation. Through our findings we infer that an efficient deployable OCR for low-resource language digitization is achievable without sacrificing practical accuracy, supporting community-driven preservation efforts.**

***Index Terms*—AI in Education, CNN-CTC, Culture for Smart Cities, Heritage Digitization, Indigenous & Low-Resource Languages, NLP, OCR, Smart Learning & Living, Sustainable AI**

I. Introduction

The Sámi languages comprise nine low-resource languages that belong to the family of Uralic languages. They are spoken in the Sápmi region, spread over the countries of Norway, Sweden, Finland, and Russia. North Sámi (Davvisámegiella) is the most widely spoken, with approximately 25,000 speakers, yet it remains classified as endangered by UNESCO. Throughout history, the governments of these countries have systematically suppressed the native population [1]. Norway has adopted the most aggressive strategy (Fornorsking[1]), banning the language, confiscating land and material goods, sending settlers to the Sápmi region and dislocating the indigenous communities. Other countries employed different discrimination techniques such as forced assimilation and relocation of indigenous people, collectivization and assertion of their material goods. Nowadays, the Nordic countries, by and large, have acknowledged their mistakes and are actively participating in the revitalization of the Sámi languages[2].

The preservation of Sámi languages now depends critically on digitizing historical documents before they deteriorate. Optical Character Recognition (OCR) is essential for this effort, yet most mainstream OCR research focuses on high-resource languages, leaving Sámi underrepresented, putting historical texts at risk. Effective OCR not only preserves heritage but also enables modern language technology applications essential for language revitalization.

North Sámi presents significant challenges for OCR systems. The scarcity of digital resources, special diacritical characters (Á/á, Č/č, Đ/đ, Ŋ/ŋ, Š/š, Ŧ/ŧ, Ž/ž), its agglutinative morphology, large vocabulary with hundreds of inflected forms per verb and a cross-linguistically rare three-way quantity contrast complicate the character recognition for Sámi orthography. Figure 1 illustrates these challenges with an example sentence from Johan Turi's 1910 work and seven special diacritical characters.

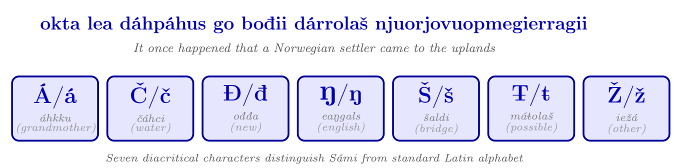


Fig. 1: North Sámi text characteristics: (a) Example sentence from Johan Turi's "Muitalus sámiid birra" [2] with English translation, and (b) Seven special diacritical characters with example words that distinguish Sámi orthography from standard Latin script.

Current state-of-the-art (SOTA) models from the National Library of Norway utilize transformer-based architecture [3] leveraging pre-training on large-scale datasets, fine-tuned for Sámi languages [4]. These are computationally demanding and time-consuming (350MB+ model size with CPU inference times exceeding 12 seconds per image). This limits their applicability in resource-constrained environments such as community-driven projects or institutions with limited infrastructure. Moreover, Enstad et al. [4] found that current baseline OCR accuracy remains insufficient for downstream NLP tasks,

[1]eng. Norwegianization

[2]Norway's parliament issued a formal apology in November 2024, after the June 2023 Truth and Reconciliation report.

highlighting the need for both improved accuracy and practical deployment solutions.

In this paper, we propose a lightweight CNN-CTC model for resource-aware transcription of Sámi language text. It thus contributes to sustainable AI. It impacts AI for 21st-century education, especially with respect to culture for smart cities and ethnic plurality, by working on low-resource indigenous language texts. This research aids smart learning and smart living. It can be useful in augmenting AI systems with NLP (natural language processing) for low-resource languages, and in training next-generation robots for everyday use (e.g. grocery stores, household use) by operating on indigenous language texts. This paper thus addresses a vital question: *Can a lightweight OCR architecture approach Transformer level accuracy for North Sámi while enabling practical deployment on modest hardware?* The main contributions of this work are:

1) Proposing a CNN-CTC model for resource-aware OCR transcription of North Sámi language texts with implementation on notable benchmark data & historical texts.
2) Approaching SOTA accuracy within 3.69 pp CER across 7 TrOCR variants on 1,048 samples, while surpassing them in speed, size, and out-of-domain accuracy.
3) Impacting Sustainable AI through a lightweight CNN-CTC model (23 MB) with 87.76% character accuracy, 380× faster inference than most Transformers.
4) Adding a linguistically motivated conjunction splitting approach that improves long sentence accuracy by 7.05 pp as a proof of concept.
5) Aiding Smart Learning by an end-to-end pipeline integrating OCR & machine translation (BLEU 10.40, chrF 33.85, TER 76.30%), good for document processing.
6) Usefulness in 21st-century education, AI & NLP, and next-gen robotic systems on low-resource languages.

II. Related Work

A. OCR for Low-Resource Languages

OCR research has traditionally focused on high-resource languages with abundant training data. Recently, more focus has been put onto low-resource setting. Agarwal and Anastasopoulos [5] provide a concise survey of OCR methods for low-resource languages, highlighting transfer learning, synthetic data generation and architecture adaptation. Rijhwani et al. [6] demonstrated OCR post-correction for endangered languages using pretrained language models, while Corallo and Varde [7] addressed OCR for Amazigh (Berber), another low-resource language with distinctive orthography. The OCR4MT benchmark [8] ties OCR for 60 low-resource languages directly to downstream MT performance. For morphologically rich languages, the vocabulary explosion problem affects both OCR confidence and language model integration. Approaches include subword tokenization [9] and character-level modeling, the latter being relevant for CTC-based architectures that operate at the character level by design.

B. Neural OCR Architectures

There are two Neural Network based approaches for OCR. First, a classical approach exemplified by CRNN [10], which combines a Convolutional Neural Network (CNN) for feature extraction and Recurrent Neural Network (RNN) for sequence modeling, trained with Connectionist Temporal Classification (CTC) loss. CTC allows alignment of length-varying sequences, therefore eliminating the need for explicit segmentation. The CRNN is efficient, explainable, and end-to-end trainable. The second approach employs Transformer architecture [11] either as encoders (e.g. Vision Transformer [12]) or encoder-decoder models. TrOCR [3] by Microsoft represents the state of the art using a Vision Transformer encoder pretrained on large image datasets and a language model decoder. National Library of Norway has fine-tuned TrOCR variants for Sámi languages [4]. While achieving better accuracy, TrOCR models are computationally expensive and time-consuming (often exceeding 10 to 15sec/image on CPU and taking 350 MB).

C. Sámi Language Technology

The TartuNLP research group at University of Tartu provides a neural machine translation API [13] supporting North Sámi to English translation (used in this work's end-to-end pipeline). The National Library of Norway, via the Språkbanken initiative, released synthetic OCR training data and fine-tuned TrOCR models for Sámi [4]. The Giellatekno project at UiT The Arctic University of Norway maintains morphological analyzers, lexical entries for many languages, proofing tools and language-learning system [14] and contributed synthetic OCR training data, through the Språkbanken initiative. Despite these advances, deployment challenges remain: Transformer models computational demands limit accessibility for resource-constrained digitization projects. Our work addresses this gap on whether lightweight architectures can provide practical accuracy-efficiency trade-offs for heritage document processing.

III. North Sámi Language and Data

A. Linguistic Characteristics

The Sámi language family consists of 9 languages (Southern, Ume, Pite, Lule, Northern, Inari, Skolt, Kildin, Ter) [15]. It is an Uralic language family of the Finno-Ugric branch, sharing typological features with Finnish and Hungarian. Its orthography, standardized in 1970s, uses the Latin alphabet augmented with seven special characters: á, č, đ, ŋ, š, ŧ, and ž (and their corresponding capitals). These characters present challenges for OCR systems: the háček (ˇ) may be confused with similar diacritics, while đ and ŧ have no visual analogs in major European languages. Additionally, North Sámi's agglutinative morphology produces large type-token ratios even in modest corpora, complicating n-gram language models used in OCR post-correction and creating vocabulary challenges for character recognition systems.

B. Training Data

For training, we use the Språkbanken synthetic Sámi OCR dataset [4], available through the Hugging Face Hub. We load the dataset's *training split* (307,387 samples) and divide it into a *training subset* (276,000 images, 90%) for model optimization and a *development subset* (approximately 31,000 images, 10%) for hyperparameter tuning and early stopping. Each image is 32×800 pixels (height×width) with corresponding ground truth transcription, generated from digital text using varied fonts, sizes, and mild distortions to simulate scanned documents. Synthetic data is necessary because no publicly available, large-scale corpus of real scanned North Sámi documents with ground truth exists. Additionally, it opens a door for recreating the approach for other low-resource languages. The character vocabulary comprises 394 unique characters covering uppercase and lowercase Latin letters, Sámi-specific characters, digits, punctuation, and common symbols, plus one CTC blank token, for 395 output classes in total. This vocabulary size reflects the linguistic diversity of the corpus while remaining feasible for CTC-based decoding.

C. Test Data

Evaluation employs test sets to assess in-domain & out-of-domain generalization, distinct from training sets and unused for hyperparameter tuning / early stopping.

Språkbanken Test Subset: 1,000 synthetic line images randomly sampled from the dataset creators' separate *validation split*. This set measures in-domain accuracy on data matching the training distribution but never seen by the model, distinct from the development subset mentioned above.

Historical Book Data: 48 hand-picked, synthetically rendered sentence pairs in Johan Turi's "Muitalus sámiid birra" (An Account of the Sámi, 1910) [2], the first secular book published in North Sámi. Images were generated using varied fonts and rendering parameters to evaluate out-of-domain generalization. It tests lexical and morphological generalization to historical vocabulary, not scan quality robustness.

The complete test set comprises 1,048 samples: 1,000 from the Språkbanken held-out split plus 48 historical book samples.

IV. Methodology

A. System Overview

Our modular OCR pipeline is structured as follows: the images are preprocessed, they pass through a CNN backbone for feature extraction, followed by adaptive pooling and projection to a fixed hidden dimension. These features are then passed directly to a linear classification head and decoded via CTC loss [16]. Figure 2 illustrates this architecture. The modular design supports different backbone (SimpleCNN, VGG16, VGG19, ResNet50, with ResNet101 in the design space but gated on the ResNet50 result and not trained due to compute) and encoder (BiLSTM, Transformer, None) combinations, enabling systematic comparison of the architectures. Due to time constraints, this work only evaluates the no encoder CTC variant type (CNN $\rightarrow$ projection $\rightarrow$ CTC head).

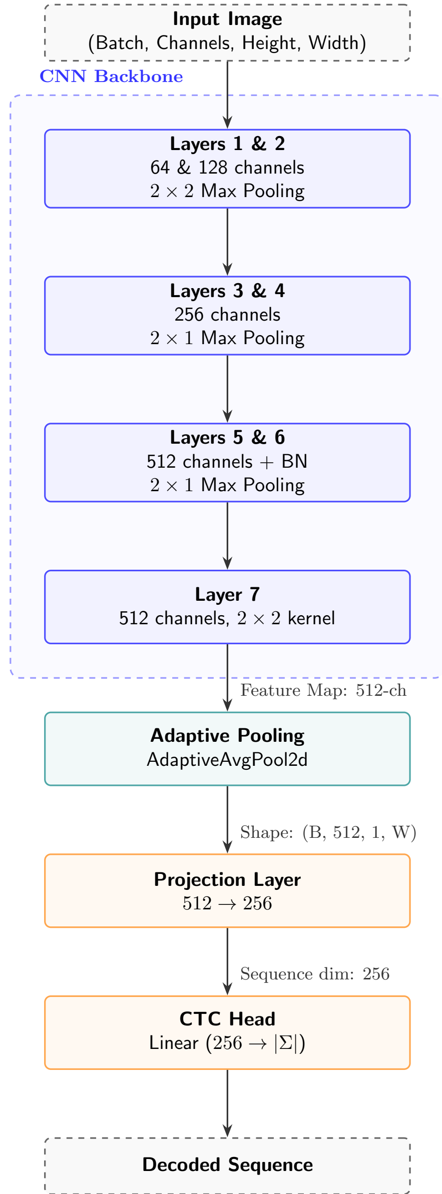


Fig. 2: Modular OCR pipeline CNN-CTC architecture. Input images (32×800 pixels) are processed through a CNN backbone, pooled to height 1, and projected to hidden dimension 256, the projected column features are passed directly to the CTC head to produce character sequences.

B. Image Preprocessing

Input images undergo grayscale conversion, aspect-ratio-preserving resize to 32 pixels height, and width normalization (padding or cropping) to 800 pixels. Pixel values are normalized from [0, 255] to [-1, 1] via $(x/255 - 0.5)/0.5$. During training, random affine transformations (rotation $\pm 2°$, translation/scale $\pm 2\%$) and ColorJitter (brightness=0.2, contrast=0.2) simulate scanning variations.

C. CNN-CTC Architecture

In line with the needs of resource-aware systems, our lightweight model (*ctc_simple*) proposes to deploy a simple

CNN backbone trained from scratch:

7 layer CNN backbone:

- Layers 1,2: 64,128 channels, 2×2 max pooling
- Layers 3,4: 256 channels, 2×1 pooling (preserved width)
- Layers 5,6: 512 channels, batch normalized, 2×1 pooling
- Layer 7: 512 channels, 2×2 kernel (height reduced to 1)

This CNN backbone is preferred due to prior success stories, e.g. it renders good results in work by Shi et al. [10].

Subsampling: MaxPool operations use stride (2,2) for early layers and (2,1) for later layers to preserve horizontal resolution while collapsing height. The output is a 512 channel feature map of reduced spatial dimensions.

Adaptive Pooling: Features are pooled to height 1 via AdaptiveAvgPool2d, yielding a tensor of shape (batch, 512, 1, *width*). *Width* depends on input width and pooling reductions.

Projection Layer: A linear transformation projects the 512 dimensional CNN features (512 backbone channels) to a hidden size of 256.

Sequence Stage: For *ctc_simple*, no recurrent or attention based encoder is applied, the projected column features are passed directly to the CTC head. This isolates the contribution of the backbone alone and yields the lightest variant of the modular pipeline.

CTC Decoder: A linear layer projects the 256 dimensional column features to 395 classes (394 characters + 1 CTC blank token), followed by LogSoftmax. During training, CTC loss marginalizes over all valid character-level alignments, at inference, greedy decoding or beam search extracts the most likely character sequence.

D. Comparison Architectures

We evaluate 11 models (7 TrOCR + 4 CTC) variants across different backbones and encoders:

Seven TrOCR variants, pretrained by the National Library of Norway [4], use a Vision Transformer encoder and GPT-2 decoder: *trocr_smi* (base Sámi only), *trocr_smi_synth* (with synthetic data), *trocr_smi_pred* (with auto-transcribed augmentation), *trocr_smi_pred_synth* (combined), and three multilingual variants incorporating Norwegian (*trocr_smi_nor*, *trocr_smi_nor_pred*, *trocr_smi_nor_pred_synth*).

Four CTC models are evaluated, our lightweight *ctc_simple* trained from scratch, plus, to test whether ImageNet pretraining benefits OCR, we also trained three models with pretrained backbones (*ctc_vgg16*, *ctc_vgg19*, *ctc_resnet50*), using the same projection layer and CTC head as *ctc_simple* (all four variants do not use the recurrent encoder). All three exhibited convergence difficulties.

E. Training Configuration

All CTC models are trained using AdamW optimization with learning rate 10^{-4} and weight decay 10^{-4} for *ctc_simple*, with reduced rates (3×10^{-5}) and differential multipliers (0.1× for pretrained layers) for VGG/ResNet variants. Batch sizes are 32 (reduced to 16 for ResNet50 due to memory constraints). Training is for up to 100 epochs with early stopping (patience 15 to 35 epochs). `ReduceLROnPlateau` halves learning rates if validation CER plateaus. Gradients are clipped to max_norm=5.0 to stabilize CTC loss. All models were trained on the Språkbanken synthetic dataset (276,000 samples). Training is on NVIDIA Tesla v100 32GB SXM2 and inference benchmarks are conducted on AMD Ryzen 5 PRO CPU to assess deployment feasibility. TrOCR models are used as provided without additional fine-tuning, representing the pre-trained baseline.

TABLE I: OCR Benchmark Comparison on 1,048 Test Samples

Model	Training Data	CER↓ (%)	WER↓ (%)	Word Acc.↑*	sec / img	Size (MB)
trocr_smi_synth	Smi+Synth	**8.55**	**28.16**	**71.84**	12.41	350
ctc_simple	Smi+Synth	12.24	37.35	62.65	**0.033**	**23**
trocr_smi_pred_synth	Smi+Pred+Synth	15.85	45.76	54.24	12.08	350
trocr_smi_nor_pred_synth	Smi+Nor+Pred+Synth	17.46	48.20	51.80	13.55	350
trocr_smi	Smi only	24.12	63.50	36.50	16.23	350

*Word Acc. = 100 - WER (word-level correctness). `smi`=Sámi data, `pred`=+auto-transcribed, `synth`=+synthetic pre-training.

F. Evaluation Metrics

Models are evaluated using standard OCR metrics:

- **Character Error Rate (CER):** Normalized Levenshtein edit distance at the character level:
$$\text{CER} = \frac{S + D + I}{N} \tag{1}$$
where S, D, I are substitutions, deletions, insertions, and N is the reference length. CER measures character-level accuracy. 0% is perfect, lower is better.
- **Word Error Rate (WER):** Edit distance computed at the word level. WER is more sensitive to errors that break word boundaries but better reflects downstream usability.
- **Exact Match and Normalized Accuracy:** Binary metric where prediction receives 1.0 if it exactly matches the reference, else 0.0. The mean across all samples gives overall exact match rate. The normalized accuracy is computed after removing punctuations and collapsing whitespaces, providing fairer assessment where the punctuation errors are not so critical.
- **Inference Time:** Seconds per image averaged over all test samples, measured on CPU to assess deployment feasibility. Includes preprocessing, forward pass, and CTC decoding but excludes batch preparation overhead.

CER and WER are computed using Levenshtein distance without Unicode normalization preprocessing. All comparisons operate on raw UTF-8 strings with no NFC/NFKC normalization applied [17]. The Språkbanken dataset uses consistent Unicode encoding.

V. Experiments and Results

Table I presents benchmark results for the top performing models (underperforming models with CER >70% omitted for clarity).

A. Main Results

According to our benchmarks, the best-performing TrOCR variant (trocr_smi_synth) achieves 91.45% character accuracy and 71.84% word accuracy, establishing the state of the art for North Sámi OCR. However, this accuracy comes at significant

TABLE II: Domain-Specific Performance: HuggingFace Validation (In-Domain) vs. Historical Book Text (Out-of-Domain)

Model	HF CER (%)	Book CER (%)	Δ CER (pp)	Book Norm Acc (%)
trocr_smi_synth	8.33	13.17	+4.84	45.8
ctc_simple	12.24	12.21	-0.03	**62.5**
trocr_smi_pred_synth	16.09	**11.00**	**-5.09**	43.8

computational cost: $\sim$ 12.41 seconds per image on CPU and a 350MB model file. Our lightweight ctc_simple model achieves 87.76% character accuracy (computed as $100-\text{CER}$), trailing the best Transformer by 3.69 pp in CER (12.24% vs 8.55%), while providing dramatic efficiency gains: 380× faster inference (0.033s vs 12.41s per image) and 15× smaller model size (23MB vs 350MB). This translates to a throughput of approximately 109,000 images per hour compared to TrOCR's 290 images per hour, a significant difference for digitization projects.

B. Generalization to Historical Text

To assess practical heritage digitization performance, we evaluate domain-specific results on synthetic HuggingFace data (in-distribution) versus historical book text (out-of-distribution). Note, that these historical test images are synthetic renderings of text from Turi's 1910 work, testing lexical and morphological generalization rather than scan quality robustness. Table II reveals a surprising finding.

While the best-performing TrOCR variant (trocr_smi_synth) maintains better performance on synthetic HuggingFace data (8.33% CER), it degrades significantly when applied to out-of-distribution historical content (13.17% CER), a degradation of 4.84 percentage points. In contrast, our lightweight CTC model exhibits unusual stability: performance on historical book text (12.21% CER) is, in fact, actually marginally better than on synthetic data (12.24% CER).

Interestingly, the trocr_smi_pred_synth variant exhibits the most dramatic domain shift improvement: from 16.09% CER on synthetic data to 11.00% CER on historical text, a 5.09pp improvement. This model achieves the best Book CER despite having the worst HuggingFace performance among the three, suggesting that the "pred" (auto-transcribed data) augmentation strategy introduces lexical patterns more representative of historical text. Consistent with Enstad et al. [4]'s observation that trocr_smi_pred_synth helps with domain shift, though small sample size limits the statistical power.

Normalized accuracy (to remove punctuation, whitespace differences) further highlights our model's practical advantage: it gives 62.5% normalized accuracy on historical text vs. 45.8% for trocr_smi_synth and 43.8% for trocr_smi_pred_synth. While the 48 sample historical test set limits statistical power, consistent patterns across metrics imply meaningful domain shift resilience. The lightweight architecture learns more generalizable character-level features rather than memorizing synthetic rendering artifacts.

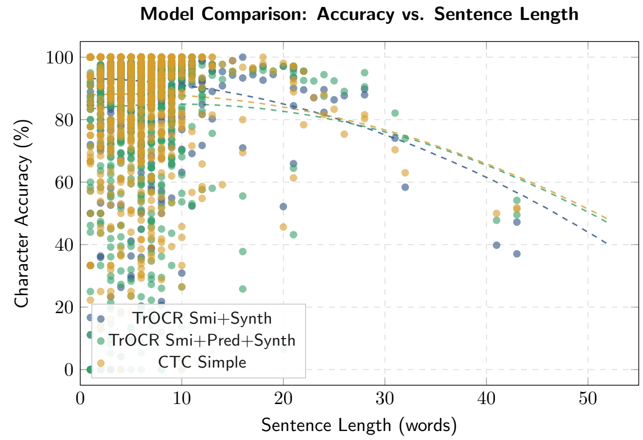


Fig. 3: Character accuracy vs. sentence length for TrOCRs (trocr_smi_synth, trocr_smi_pred_synth) and ctc_simple models. Both architectures degrade beyond ∼80 characters due to fixed 800 pixel input width compressing longer text. Points are single test samples, lines are regression trends.

C. Sentence Length Analysis

Figure 3 shows character accuracy versus sentence length for both architectures.

Both architectures exhibit degrading accuracy with increasing sentence length, with a critical threshold around 80 to 100 characters. This degradation stems from the fixed 800 pixel input width: longer sentences are compressed into the same pixel budget, reducing effective resolution per character.

For sentences under 80 characters, TrOCR(trocr_smi_synth) maintains higher accuracy but with greater variance. The CTC model shows more consistent performance across all lengths, likely due to its simpler architecture being less sensitive to input compression artifacts.

Conjunction-Based Splitting Mitigation: To address this length-based degradation, we implement a linguistically motivated preprocessing approach: splitting long sentences at North Sámi conjunction boundaries before OCR, then rejoining predictions. North Sámi uses coordinate conjunctions (*ja* "and", *muhto* "but", *dahje* "or") and subordinate conjunctions (*go* "when/because", *ahte* "that", *jos* "if") [18] that naturally delimit clause boundaries. Only texts exceeding 80 characters are processed, splitting *before* conjunctions, maintaining minimum segment lengths of 20 characters to avoid over-splitting.

Figure 4 shows the results. For long sentences (>80 characters, n=67), conjunction-based splitting dramatically reduces CER from 15.17% to 8.12%, an improvement of 7.05 percentage points. What is noteworthy, is that this brings long sentence CER *below* that of short sentences (8.12% vs 12.04%), effectively eliminating the length penalty. A paired t-test confirms statistical significance ($p = 0.000009$) with a large effect size (Cohen's $d = 1.35$). Critically, 95% of split samples showed improvement with no degradation cases. Because the splitter operates on ground truth, this is an oracle upper bound: a deployed system would have to detect conjunctions on noisy OCR output, so only a fraction of the gain would carry over. We therefore present conjunction

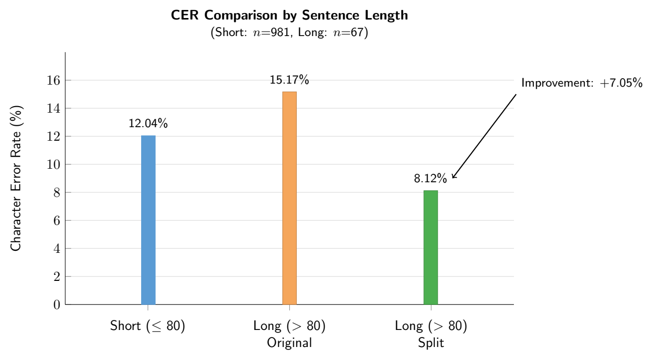


Fig. 4: CER comparison showing the effect of conjunction-based splitting. Long sentences (>80 chars) degrade to 15.17% CER when processed whole, but splitting at conjunction boundaries reduces error to 8.12%, even better than short sentences (12.04%).

TABLE III: Split Example: Conjunction-Based Segmentation

Ground Truth (137 chars)
Ja dasa lea dát sivva: ***go*** *sápmelaš boahtá moskkus gámmirii, de son ii ipmir ii báljo maidege,* ***go*** *ii biegga beasa bossut njuni vuostá.*

Original Prediction (19.7% CER)
Ja dasa lea dát sivva: go sápmelaš boahtá moskkus gámmirii, de son ii ipmir ii báljo maidege, go ii biegga bea

Split Processing → Rejoined (0.0% CER)
Seg. 1: *Ja dasa lea dát sivva: go sápmelaš boahtá moskkus gámmirii, de son ii ipmir ii báljo maidege,*
Seg. 2: *go ii biegga beasa bossut njuni vuostá.*
Joined↪ *Ja dasa lea dát sivva: go sápmelaš boahtá moskkus gámmirii, de son ii ipmir ii báljo maidege, go ii biegga beasa bossut njuni vuostá.*

Split at second *go* conjunction, each segment fits within 800px width constraint.

splitting as a proof of concept.

Table III illustrates this improvement with a concrete example from Turi's work. A 137-character sentence, when processed whole, truncates after ∼110 characters (19.7% CER). The splitter detects a subordinate conjunction *go* ("when/because") and divides the input in 2 segments, each processed independently, then rejoined to achieve 0% CER.

This finding suggests that simple, linguistically informed preprocessing can substantially mitigate the length constraint without architectural changes. We hypothesize this approach generalizes to other agglutinative languages with similar clause-marking conjunctions.

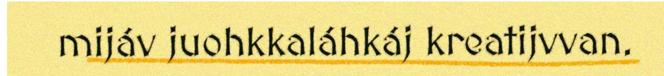
D. End-to-End Pipeline Evaluation

To demonstrate practical applicability, we evaluated a proof-of-concept OCR→MT pipeline using the TartuNLP neural machine translation API [13] to translate predicted North Sámi text to English. Using *ctc_simple* OCR outputs, the pipeline achieved BLEU [19] 10.40, chrF [20] 33.85, and TER 76.30%.

These moderate translation scores reflect the cascading effect of OCR errors: a 12.24% character error rate propagates through to mistranslations and reduced fluency. Qualitative analysis shows that OCR errors in morphological suffixes (e.g., case markers) disproportionately affect MT output, as the agglutinative structure means single character errors can change grammatical relations. Additionally, this highlights the need for reliable, open-access translation tools bridging high-resource languages such as English and low-resource languages like Sámi. Nevertheless, the system can process historical documents end-to-end without manual transcription, providing rough translations suitable for archival indexing.

TABLE IV: OCR Recognition Examples with Ground Truth Images (ctc_simple model)

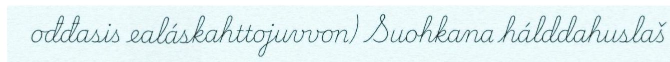
Correct Recognition (CER = 0%)



Ground Truth: *mijáv juohkkaláhkáj kreatijvvan.*
Prediction: *mijáv juohkkaláhkáj kreatijvvan.*

Ground Truth: *julev- ja davvisámigielagiiguin ja sámiiguin geat eai*
Prediction: *julev- ja davvisámigielagiiguin ja sámiiguin geat eai*

Diacritical Errors (CER = 8.3%)



Ground Truth: *ođđasis ealáskahttojuvvon) Suohkana hálddahuslaš*
Prediction: *oddasis ealáskahttojuvvon) Duohkama hálddahuslaš*
Note: ođđasis → oddasis shows đ→d error (stroke missed)

Uppercase Text Errors (CER = 29.6%)

Ground Truth: *FYLKKAGIELDDAIGUIN. FINNMÁRKKU FYLKKAGIELDDAS LEA ÁIGUMUŠŠIEHTADUS SÁMI*
Prediction: *FYLKKAGIELDDAIGUI0. F1O0mÁRKKU FYLKKAGIELDDAs LEA ÁIGUmUŠe*

E. Error Analysis

Qualitative examination of prediction errors reveals several patterns, illustrated in Table IV.

ImageNet-Pretrained Models: The three CTC models using ImageNet-pretrained backbones (ctc_vgg16: 73.9% CER, ctc_vgg19: 71.9% CER, ctc_resnet50: 75.3% CER) did not converge to usable performance despite using the same projection layer and CTC head as ctc_simple. All three models achieved 0% exact match accuracy, producing largely unintelligible character sequences.

We hypothesize that ImageNet's object-level features (edges, textures, boundaries) misalign with character-level recognition (stroke topology, diacritics, spacing), while SimpleCNN trained from scratch learns task-appropriate features. The 70 to 75% vs 12.24% CER gap points to a representation problem rather than under optimization, though this conclusion holds only under our compute budget.

Consonant Clusters: The CTC model occasionally drops characters in complex consonant clusters common in Sámi morphology (e.g., predicting "sggoda" for "cuiggoda"). TrOCR handles these better, likely due to its decoder's language modeling capacity. Post-correction integration could mitigate this issue (See Section VII).

Diacritical Character Handling: Table V presents per-character accuracy for the seven Sámi-specific diacritics. TrOCR(trocr_smi_synth) achieves 93.4% overall diacritic accuracy compared to the CTC model's 88.3%. Both models struggle most with ŧ (50% accuracy), likely due to its visual similarity to standard 't' and extreme rarity in the test set (only 6 occurrences). Interestingly, the CTC model achieves perfect accuracy on ŋ (26 occurrences) while TrOCR confuses it with 'g' (84.6%). The high accuracy on á (90 to 96%) benefits from its frequency (1,473 occurrences). Common error patterns include á→a diacritic stripping, đ→d confusion (the stroke being missed), and case confusions (e.g., Á↔á).

Case Sensitivity: Both models struggle with all uppercase text (a small part of training samples). This suggests a need for case-augmented training or case normalization preprocessing.

Length-Dependent Errors: As shown in Figure 3, both models degrade on sentences exceeding 80 characters due to the fixed 800 pixel input width constraint discussed in Section IV-B with mitigation technique in Section V-C.

TABLE V: Per-Character Accuracy for Sámi Diacritics

Character	Count	ctc_simple	trocr_smi_synth
á / Á	1473	90.1%	95.9%
č / Č	129	83.7%	81.4%
đ / Đ	191	79.1%	87.4%
ŋ / Ŋ	26	100.0%	84.6%
š / Š	262	87.8%	93.1%
ŧ / Ŧ	6	50.0%	50.0%
ž / Ž	35	80.0%	85.7%
Overall	2122	88.3%	93.4%

VI. Discussion

A. Practical Deployment Considerations

Our lightweight CNN-CTC model paves the way for deployment scenarios where TrOCR remains impractical. At 109,000 images per hour, a single CPU can process a million pages in under 10 hours vs. 144 days for TrOCR, allowing cultural institutions to complete Sámi digitization projects with reasonable time and budget. The 23MB footprint permits deployment on mobile devices for field documentation in remote areas, supporting indigenous data sovereignty by avoiding cloud-based processing. Latency below 100ms enables interactive transcription interfaces with immediate feedback.

While TrOCR may be preferable for cases where precision is paramount, the 3.69pp CER gap represents an acceptable trade off for bulk indexing and content discovery. Our model's stable performance on synthetically rendered historical lexical content (Section V-B) suggests practical reliability may exceed what synthetic benchmarks alone indicate.

B. Limitations

There are several limitations constraining the present work. All models were trained on synthetic data rather than real scans, while historical evaluation (Section V-B) demonstrates lexical generalization, performance on physical degradation (faded ink, paper damage) remains untested and requires a corpus of real scanned Sámi documents. The pre-segmentation of text lines is assumed, which requires layout analysis tool integration for full-page processing, and the fixed 800 pixel OCR width constrains very long lines. Additionally, our models perform pure vision-based OCR without morphological post-correction. Integrating Giellatekno's analyzer [14] could improve accuracy for rare inflected forms.

We did not evaluate classical OCR baselines (Tesseract, Transkribus), our focus was on comparing lightweight neural architectures against state-of-the-art Transformer models. Enstad et al. [4] provides Tesseract comparisons showing substantially higher CER (>30%) on Sámi text.

C. Ethical Considerations

This work addresses language technology for a historically marginalized community. Following the years of policies that suppressed Sámi languages, revitalization is a matter of cultural survival and historical justice. Technology must support, not replace, community efforts. Our edge-deployable model enables local processing without uploading sensitive materials to external servers. We provide our work results to benefit institutions and communities, highlighting the need to prevent commercial exploitation of indigenous language resources. Following the CARE Principles for Indigenous Data Governance [21], emphasizing Collective Benefit, Authority to Control, Responsibility, and Ethics, we commit to releasing models and code under the MIT license via GitHub and HuggingFace. All source material consists of published historical texts in the public domain. Formal Free, Prior and Informed Consent (FPIC) was not required as no unpublished community materials were used.

VII. Conclusion and Future Work

This work demonstrates that lightweight CNN-CTC architectures offer viable, high-efficiency alternatives to Transformer-based architectures for North Sámi OCR. Our `ctc_simple` model achieves 12.24% CER, only 3.69pp behind the best TrOCR variant, while operating 380× faster with a 15× smaller footprint, enabling high-throughput (109,000 images/hour on CPU), making large-scale digitization feasible. Notably, our lightweight model exhibits stable performance on 1910 historical text (n =48) while TrOCR degrades by 4.84pp, suggesting that smaller architectures may generalize better to domain-specific features than Transformers. We find that pretrained generic ImageNet backbones fail to converge in this task, indicating that training a model from scratch may better suit specialized indigenous language recognition. Linguistically motivated sentence splitting mitigates length-based degradation under oracle conditions ($p < 0.00001$), indicating a promising preprocessing direction that remains to be validated on predicted text.

Figure 5 summarizes targeted applications of this work. In the future, combining a diffusion model and existing OCR so-

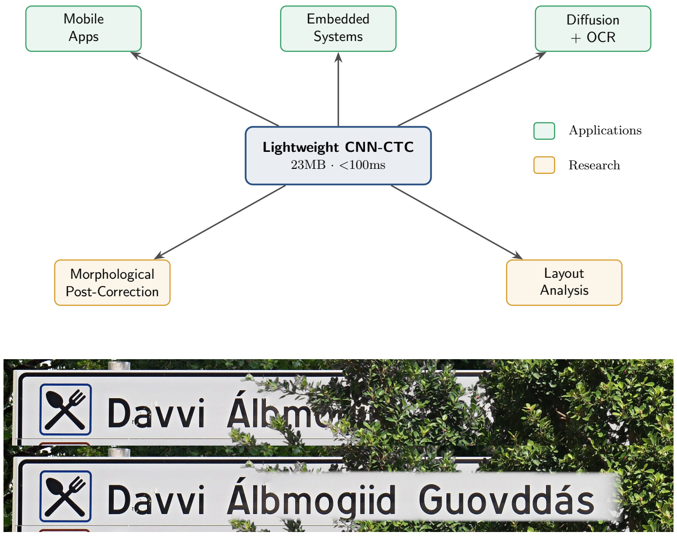


Fig. 5: (a) Targeted applications and future directions for our lightweight OCR model, (b) Potential use case in damaged text reconstruction, e.g. in autonomous driving (this can also be useful for grocery store labels etc.)

lution can yield unified pipelines for damaged text reconstruction. Through small size (23MB) and low latency (< 100ms), this is deployable on embedded devices useful in self-driving vehicles when road signs are occluded or organizing of shop items with damaged labels. Our model's portability makes it suitable for mobile applications in document digitization, useful in areas without internet connectivity. To improve accuracy, especially for rare inflected forms, a Giellatekno's morphological analyzer [14] for post-correction can be integrated, though it needs corpora of real scanned documents to train authentic degradation artifacts. Line-level OCR can be combined with layout analysis (de-skewing, line detection) to enable full-page document processing. Future work can also entail implementation of some targeted applications, especially to train next-gen robots. Further related exploration can be in languages with challenges of fully pictorial scripts [22] and in aspects such as cultural commonsense [23] as well as human-robot collaboration using text & images [24], possibly collating with earlier work by our research teams.

Our research in this paper is noteworthy for impacts on 21st-century education by addressing resource-aware OCR in indigenous low-resource languages. It aids smart learning & smart living due to the usefulness of the OCR transcription and its targeted applications.

Acknowledgments

We thank the National Library of Norway (Språkbanken) for providing pre-trained TrOCR models and synthetic training data. We acknowledge Giellatekno (UiT The Arctic University of Norway) for the SIKOR corpus and test data. We thank TartuNLP (University of Tartu) for providing the North Sámi to English translation API. Computational resources were provided by the University of Southern Denmark. Aparna Varde acknowledges a grant from NNF, Denmark, in the area of Sustainable AI.

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